



Invited Review Article

Advances in the pathophysiology, diagnosis and management of chronic diarrhoea from bile acid malabsorption: a systematic review

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ABSTRACT

Bile acid malabsorption (BAM) is an important disorder of digestive pathophysiology as it generates chronic diarrhoea. This condition originates from intricate pathways involving bile acid synthesis and metabolism in the liver and gut, the composition of gut microbiota, enterohepatic circulation and key receptors as farnesoid X receptor (FXR), fibroblast growth factor receptor 4 (FGFR4), and the G-protein bile acid receptor-1 (GPBAR-1). Although symptoms can resemble those related to disorders of gut brain interaction, accurate diagnosis of BAM may greatly benefit the patient. The empiric diagnosis of BAM is primarily based on the clinical response to bile acid sequestrants. Specific tests including the 48-hour fecal bile acid test, serum levels of 7 α -hydroxy-4-cholesten-3-one (C4) and fibroblast growth factor 19 (FGF19), and the ⁷⁵Selenium HomotauroCholic Acid Test (SeHCAT) are not widely available. Nevertheless, lack of diagnostic standardization of BAM may account for poor recognition and delayed management. Beyond bile acid sequestrants, therapeutic approaches include the use of FXR agonists, FGF19 analogues, glucagon-like peptide-1 (GLP-1) receptor agonists, and microbiota modulation. These novel agents can best make their foray into the therapeutic armamentarium if BAM does not remain a diagnosis of exclusion. Ignoring BAM as a specific condition may continue to contribute to increased healthcare costs and reduced quality of life. Here, we aim to provide a comprehensive review of the pathophysiology, diagnosis, and management of BAM.

1. Introduction

Excessive biosynthesis or secretion of bile acids (BAs) and their malabsorption (BAM) in the terminal ileum or the colon [1] generates chronic diarrhoea (cholorrhea). This symptom depends on a dysregulation in the enterohepatic circulation of BAs (Fig. 1) and, in turn, on relative abundance of BAs in the lower gastrointestinal tract, with abnormal transport of water and sodium, mucosal damage, increased mucus secretion and intestinal motility, gut dysbiosis [2,3]. Symptoms like urge of defecation [4–7], fecal incontinence [8], fatigue, dizziness, and fainting feeling [9] in BAM may lead to depression and poor quality of life [8,10]. Four subtypes of BAM are linked to the following etiologies: ileal disease (Type I), idiopathic BAM (Type II), non-ileal conditions leading to BAM (Type III), and excessive BA synthesis (Type IV)

[10] (Table 1).

A diagnostic delay is common, and up to one-half of patients with BAM report symptoms lasting for more than 5 years before the correct diagnosis [8]. Accordingly, timely and accurate identification of BAM is a major unmet need.

Here, we discuss relevant aspects relative to diagnosis and management of BAM.

2. Methods

A systematic literature search was conducted in PubMed to identify studies on BAM published from 1965 to May 19, 2024. The search used a combination of keywords and medical subject headings (MeSH) related to BAM: (“bile acid malabsorption”[MeSH Terms] OR “bile acid diarrhoea”

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OR "bile salt malabsorption" OR "bile salt diarrhoea" OR BAM) AND ("diarrhoea"[MeSH Terms] OR "chronic diarrhoea" OR "malabsorptive diarrhoea").

Studies were included (see flow-chart in Fig. 2) if: (1) focused on BAM and its association with BA-induced diarrhoea; (2) provided original data. Reviews, editorials, and letters were excluded unless they

contained relevant data. Selected articles were also analysed to identify cited studies that may have been missed during the initial search.

3. What is the prevalence of BAM and who is at risk?

In a US population study on patients referred to a gastroenterology

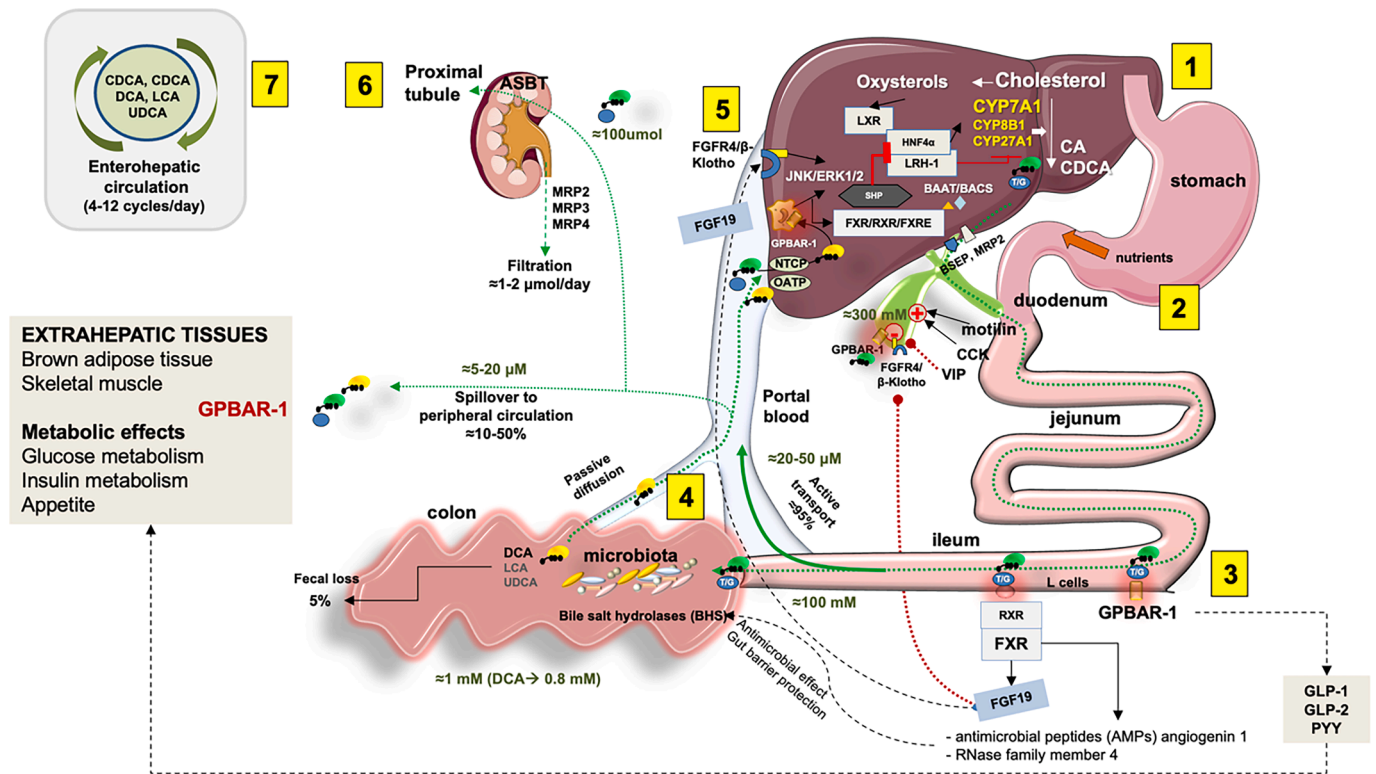


Fig. 1. Bile acid (BA) biosynthesis, enterohepatic circulation and function through their receptors and enterohormones in the liver and intestine. 1) Liver: primary BAs cholic (CA) and chenodeoxycholic acid (CDCA) are mainly synthesized from cholesterol by cholesterol 7α -hydroxylase (CYP7A1) and by CYP8B1 at a later step (the “classical pathway”), and by the CYP27A1 (the “alternative pathway”). BAs are then conjugated to taurine or glycine mainly via two enzymes, the BA CoA synthase (BACS) and the BA-CoA-amino acid N-acetyltransferase (BAAT), secreted into bile by the bile salt export pump (BSEP). The multidrug resistance-associated protein (MRP2) mediates the secretion of organic substrates such as bilirubin, and glutathione. 2) Gallbladder: bile is stored, concentrated and periodically secreted in the duodenum during gallbladder contraction. In the fasting period $\sim 20\%$ emptying occurs at the end of phase II of the migrating myoelectric complex [150,151] under the control of the vagus and enterohormone motilin [151]. In the postprandial period $>50\%$ emptying occurs due to fat-induced release of the enterohormone cholecystokinin (CCK) [152]. This rhythmic activity is also modulated in concert with episodes of gallbladder relaxation, i.e., refilling due to the inhibitory effect of other enterohormones. The vasointestinal peptide (VIP) is released in the duodenum by the gastric acid. The BAs/retinoid X receptor (RXR)/farnesoid X receptor (FXR) interaction in the terminal ileum activates the release of the human fibroblast growth factor 19 (FGF19) which activates the FGF receptor 4 (FGFR4)/ β -Klotho receptor with inhibitory effect on gallbladder contraction [120,152,153]. The BAs-dependent stimulation of the G protein-coupled bile acid receptor 1 (GPBAR-1)-cAMP-PKA signalling cascade in gallbladder smooth muscle cells leads to the opening of ATP-regulated potassium channels and subsequent muscle cell relaxation and gallbladder filling [154]. Fasting serum BAs concentrations in healthy subjects are $0.2\text{--}0.7\mu\text{M}$ to increase to $4\text{--}5\mu\text{M}$ after each meal [155–158]. 3) Terminal ileum: BAs activate FXR and increase the transcription of the enterokine FGF19 in humans (or FGF15 in mice) which enters the portal circulation and regulates both gallbladder (see point 2) and liver effects (see point 5). BAs in the intestine also activate the GPBAR-1 and stimulate the secretion of peptide YY (PYY), glucagon-like peptide 1 (GLP-1) and glucagon-like peptide 2 (GLP-2), all of which produce important metabolic effects on glucose metabolism [159] insulin metabolism and appetite acting on GPBAR-1 receptors located in the cells of brown adipose tissue and muscle [159,160]. 4) Terminal ileum-colon: BAs are efficiently (i.e., $>95\%$) reabsorbed in the terminal ileum. The remaining BAs enter the colon, undergo biotransformation to the secondary BAs by the resident gut microbiota, and undergo passive diffusion and reabsorption. Only 5% of BAs are lost in feces daily. The enterohepatic circulation of BAs includes their intestinal re-absorption and continuous recirculation to the liver through the portal vein. About $10\text{--}50\%$ of re-absorbed BAs undergo peripheral spillover into systemic circulation [161]. 5) Liver: a set of nuclear receptors, i.e., FXR, RXR, small heterodimer partner (SHP), liver receptor homologous-1 (LRH-1), and liver X receptor (LXR) [162] play a key role in the liver. The circulating FGF19 binds to hepatic FGFR4/ β -Klotho to activate c-Jun N-terminal kinase/extracellular signal-regulated kinase (JNK/ERK) signalling [163], which inhibits expression of CYP7A1 and CYP8B1 and hepatic BA synthesis, in synergy with the FXR-SHP inhibitory pathway [164,165]. BAs enter the liver by sodium taurocholate cotransporting polypeptide (NTCP) and organic anion transporting polypeptide (OATP) transporters and act as physiological nuclear ligands for FXR, which regulates target gene transcription by binding to RXRs as a heterodimer [166]. This results in increased transcription of the SHP expression. SHP, in turn, inhibits LRH-1, preventing the activation of target genes that participate in BAs and fatty acid synthesis. In the absence of BAs, LRH-1 acts together with LXR to stimulate BA synthesis [167–169]. FXR also regulates the enzymatic activity that is involved in BA conjugation to glycine or taurine, and hepatic BA secretion by BSEP and hepatic phospholipid secretion by ABCB4. BAs re-entering the liver also interact with the liver GPBAR-1 expressed in Kupffer cells, in concert with the pathway activated by the FGFR4/ β -Klotho. FXR activation also coordinates BA detoxification enzymes (i.e., cytosolic sulfotransferase 2A1 (SULT2A1), aldol-keto reductase 1 B7 (AKR1B7), cytochrome P450 3A4/3A11 (CYP3A4/Cyp3a11), and UDP-glycosyltransferase 2B4 (UTG2B4)) [170]. 6) Kidney: from the peripheral circulation, BAs also undergo renal uptake by the apical sodium/dependent BA transporter (ASBT) in the proximal tubule. Glomerular filtration of BAs is regulated by MRP 2, 3, 4 transporters [171]. 7) Enterohepatic circulation of BAs: consists of 4–12 cycles daily of both primary, secondary, and tertiary conjugated and unconjugated BAs.

Table 1
Classification of BAM.

Type	Main etiology	Notes
Type 1	Defective BA reabsorption due to ileal disease [88–90].	• Frequently seen in Crohn’s disease with (>90%) or without resection of the ileum (11–76%) [62, 91]. • May occur in patients undergoing ileal resection for any reason and due to radiation enteropathy [92]. • Can manifest as osmotic diarrhoea or true steatorrhea secondary to fat malabsorption [10].
Type 2	Idiopathic (primary) BA malabsorption	• Defects in BA absorption is not documented. • May present as primary BA diarrhoea or diarrheal variant of IBS (IBS-D) [32,93] • Associated with reduced microbial biotransformation of primary to secondary BAs • Genetic variants responsible for rare (~1%) primary BAM [94]. An association between the rs3808607 T>G polymorphism, located within the promoter region of CYP7A1, and elevated 7α-hydroxy-4-cholesten-3-one (C4)-to-cholesterol ratios exists in patients developing secondary BAD after ileal resection surgery [95]. Likely, a twofold increase in BAs synthesis occurs among individuals carrying the TT (AA) genotype. CYP7A1 polymorphisms relates to both fecal primary BAs levels and colorectal adenoma risk found with heterozygosity for the rs8192877 A>G variant, since individuals exhibited approximately double the fecal primary BAs compared to those homozygous for AA, and even higher values occurred in GG homozygotes [96]. Other variants may lead to altered ASBT [97], reduced FGF19 synthesis and deficient inhibition of BA synthesis in the liver [98]. The key role of the hepatocyte FGF19/FGR4-β-Klotho/ERK1/2/CYP7A1 axis [99,100] must be taken into account as well since it could be dysregulated due to genetic polymorphisms of FGR4-β-Klotho variations and dysregulated FGF19 response, with significant associations with primary BAD and IBS-D [98, 101], due to increased BA synthesis [102]. Animal models [103,104] point to the role of these pathways in a subgroup of individuals with BAD, and notably impaired release of ileal FGF19 occurs in cholestyramine-responsive IBS-D patients, suggesting that excessive hepatic BAs synthesis can indeed depend on disrupted FGF19 signaling [105]. Genetic variants of GPBAR-1 SNP rs11554825 (minor allele frequency 41%) were correlated with symptoms in a study comparing scintigraphically-assessed small bowel and colonic transit of 230 healthy controls and 414 patients with functional gastrointestinal disorders (84 IBS alternators, 157 IBS-C, 173 IBS-D) [106]. The role of gut microbiota in type 2 BAM requires further studies. • In patients with chronic watery diarrhoea Type 1 BAM prevalence by 75SeHCAT was 34% [62]. • Prevalence of type 2 BAM estimated in few studies and is as low as 1% of new cases of patients with chronic diarrhoea and 3% of patients with overall chronic diarrhoea, suggesting underdiagnosing [107]. Nevertheless, 51% of patients with unexplained diarrhoea undergoing the 48-h fecal BAs testing had BAM. About two-third of them improved after BAs sequestrant therapy but only one-third if BAM was absent [108]. In patients with chronic diarrhoea tested by 75SeHCAT, BAM was detected in 38% of cases and Type 2 in 28% of cases [109]. Others reported that BAM tested by 75SeHCAT is found in 50% of patients and one-third have IBS-D [28]. A systematic review estimated the population prevalence of BAM

Table 1 (continued)

Type	Main etiology	Notes
		>1% and described BAM tested by 75SeHCAT in about 30% of patients with chronic diarrhoea. The results of 15 prospective studies on patients with IBS-D-like symptoms, the prevalence of BAM by 75SeHCAT was 10%, either moderate (32%) or mild (26%). Response rate to cholestyramine was 96% and 70% in patients with severe or mild BAM, respectively [12]. About one-fourth of IBS-D [110] or functional diarrhoea patients [21] are positive for BAD. Other studies reported the association between BAM and post-infective diarrhoea, i.e., 55% of patients with a history of acute gastroenteritis and positive 75SeHCAT test [111] with 18% of patients with post-infective diarrhoea and BAM responsive to cholestyramine [112]. Fecal levels of primary unconjugated BAs and serum C4 were high in patients with IBS-D secreting more BAs at higher body size, but lower in IBS with constipation (IBS-C) or healthy volunteers [113]. There is a role for BAs in the pathogenesis of IBS, and potential biomarkers of BAM can be serum C4 and primary and secondary unconjugated fecal BAs [38].
Type 3	Non-ileal conditions leading to BA malabsorption	• In patients with chronic watery diarrhoea Type 3 BAM prevalence by 75SeHCAT was 28% [62]. • One example is post-cholecystectomy diarrhoea observed during the same year of cholecystectomy [62] due to post-prandial release of a more continuous drip of less concentrated BAs [114, 115]. Cholecystectomized individuals show about doubled recycling of the BA pool through the intestine [116,117]. This accelerated recycling of BAs is associated to a twofold increase in BA synthesis in mice and humans [118] and increased BA and cholesterol biliary secretion rates [119–121]. Although the BA pool remains constant, fat is digested and absorbed normally in cholecystectomized individuals [122]. In this condition, gut bacteria come in contact with increased exposure of primary BAs which undergo anaerobic bacteria deconjugation and dehydroxylation to secondary BAs, mainly DCA [120,123,124]. The BAs pool becomes enriched in more hydrophobic, less hydrophilic BAs with a number of toxic effects and consequences. For example, DCA can disrupt cell membranes leading to cell lysis [125,126]. Clinical studies have previously shown that in the short-term after cholecystectomy, CA synthesis slightly decreases but the pathway CA $\rightarrow$ 7α-dehydroxylation $\rightarrow$ DCA increases with more recirculating DCA to the liver and a negative feedback on hepatic CA synthesis [127]. The change in BAs profile does not change on the long-term after cholecystectomy [128]. In addition, fecal BAs excretion increases after cholecystectomy in both mice and humans [114,115,123,124, 129–131]. The incidence of diarrhoea after cholecystectomy has a wide range, from 2% to 50% depending on populations, study design, and characteristics of diarrhoea [24,64,132]. The diarrhoea usually resolves or improves over the course of weeks to months post-cholecystectomy [24]. According to other authors, BAM incidence is 68–86% after cholecystectomy [28,62], but only 9.1% in patients with diarrhoea after cholecystectomy (two-thirds of them diagnosed with BAM) in a larger analysis of 25 studies [133]. • Other causes of type 3 BAM include post-vagotomy diarrhoea, chronic pancreatitis with impaired bicarbonate secretion [134], celiac disease due to villous atrophy, and impaired gallbladder and small bowel motor function

(continued on next page)

Table 1 (continued)

Type	Main etiology	Notes
Type 4	Excessive BA synthesis without primary malabsorption	[135], although the prevalence of BAM was comparable to that of healthy controls [28,109].
		- BAM in microscopic colitis depends on villous atrophy, inflammation and collagen deposition in the ileum paving the way to BA malabsorption [136]. ⁷⁵SeHCAT retention < 10% was found in about 44% of patients presenting with collagenous colitis and chronic diarrhoea and almost 80% were responsive to BAs sequestrant therapy [137], suggesting therapy in all patients with microscopic colitis despite a negative ⁷⁵SeHCAT test [138]. - Essentially occurs in hypertriglyceridemia and during antidiabetic therapy with metformin [139–141]. - About 20% of patients on metformin had accelerated bowel transit times [142]. - Increased hepatic BAs production and diarrhoea occurs in patients with metabolic dysfunction-associated steatotic liver disease (MASLD) [143, 144], who also display elevated serum C4 [145]. - Patients with obesity and hypertriglyceridemia can show increased BAs synthesis and idiopathic BAM [49,146].

outpatient clinic, the prevalence of chronic diarrhoea was about 1% [11], and the prevalence of BAM among patients with unexplained chronic diarrhoea is about 25% to 35% [12]. In addition, about 30% of patients treated for diarrhoea-predominant irritable bowel syndrome (IBS-D) suffer from idiopathic BAM [13].

The prevalence of BAM is higher in patients at risk for this condition, as those undergoing ileal resection [14] or with terminal ileal diseases (e.g., Crohn disease, radiation-induced injury). In addition, BAM may occur frequently in gastrointestinal diseases associated with impaired motility or absorption (as seen in celiac disease) or with dysregulated delivery of BAs to the small intestines, as seen following cholecystectomy [15]. BAM may also contribute to post-vagotomy diarrhoea in patients who show higher fecal BA excretion [16,17]. Between 2003 and 2021, a Danish study identified 5264 individuals affected by BAM based on ⁷⁵Selenium HomotauroCholic Acid Test (⁷⁵SeHCAT). Compared to age and sex-matched controls from the general population, those in the BAM cohort exhibited higher prevalence of comorbidities, increased healthcare utilization, as well as lower levels of education and income [18].

4. Do we have specific clinical and diagnostic tests for BAM?

BAM and related diarrhoea is a frequently underdiagnosed condition and an early diagnosis is essential to avoid unnecessary diagnostic tests and to improve the quality of life [19,20]. While all available tests have limitations, they can help identify cases that will mostly benefit from treatment [21]. These tests are rarely used in combination due to their limited availability and cost. Diagnostic/therapeutic aspects appear in Fig. 3, while sensitivity and specificity of different diagnostic tests are discussed in Table 2.

4.1. Empiric use of bile acid sequestrants

This approach is recommended in patients with unexplained chronic diarrhoea when other specific tests are not available [22]. Studies comparing the accuracy of the empiric vs. specific testing are missing. The most important limitation is the scarce compliance of patients, with false-negative results possibly depending on the poor palatability of BA sequestrants [23].

4.2. ⁷⁵SeHCAT

This is the most accurate test to identify the presence and severity of BAM, which relies on the assessment of the abdominal retention of BAs 7 days after ingestion [24]. ⁷⁵SeHCAT is based on administration of the synthetic conjugated BA homotaurocholic acid labelled with ⁷⁵selenium (Se) isotope. Homotaurocholic is excreted by the liver and reabsorbed in the terminal ileum [25]. The patient must avoid BAs sequestrant therapy for a few days and fast for 4 hours before ingesting the ⁷⁵SeHCAT capsule [26]. The gamma camera takes the first scan within 3 hours from ingestion, and the second scan after 7 days, rendering the test less dependent on daily dietary effects [27]. Fecal BA loss is indicated by low retention of ⁷⁵SeHCAT [28]. BAM is classified as severe, moderate, or mild based on the percentage of retained radioactive selenium (<5%, 5–10%, and 10–15%, respectively) [12]. In a systematic review of 43 studies enrolling 1,223 IBS patients, <5%, <10% and <15% retention thresholds were found in 10%, 27%, and 13% of the cohort, respectively [12]. The test involves limited radiation exposure. ⁷⁵SeHCAT activity is small (370 KBq) with dose of radiation of 0.26 mSv and a total absorbed radiation of 0.3 Gy/kBq [29]. Despite the high diagnostic accuracy, ⁷⁵SeHCAT is not widely available and not accessible for most clinicians [30]. ⁷⁵SeHCAT requires a nuclear medicine department with trained personnel and is time-consuming for the patient [31]. These limitations

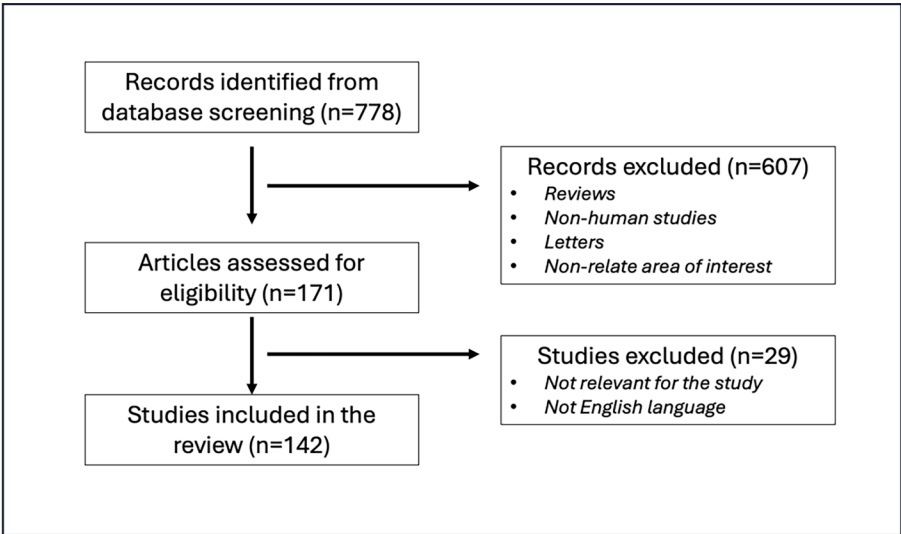


Fig. 2. Flow-chart of search strategy with final number of studied included in the review.

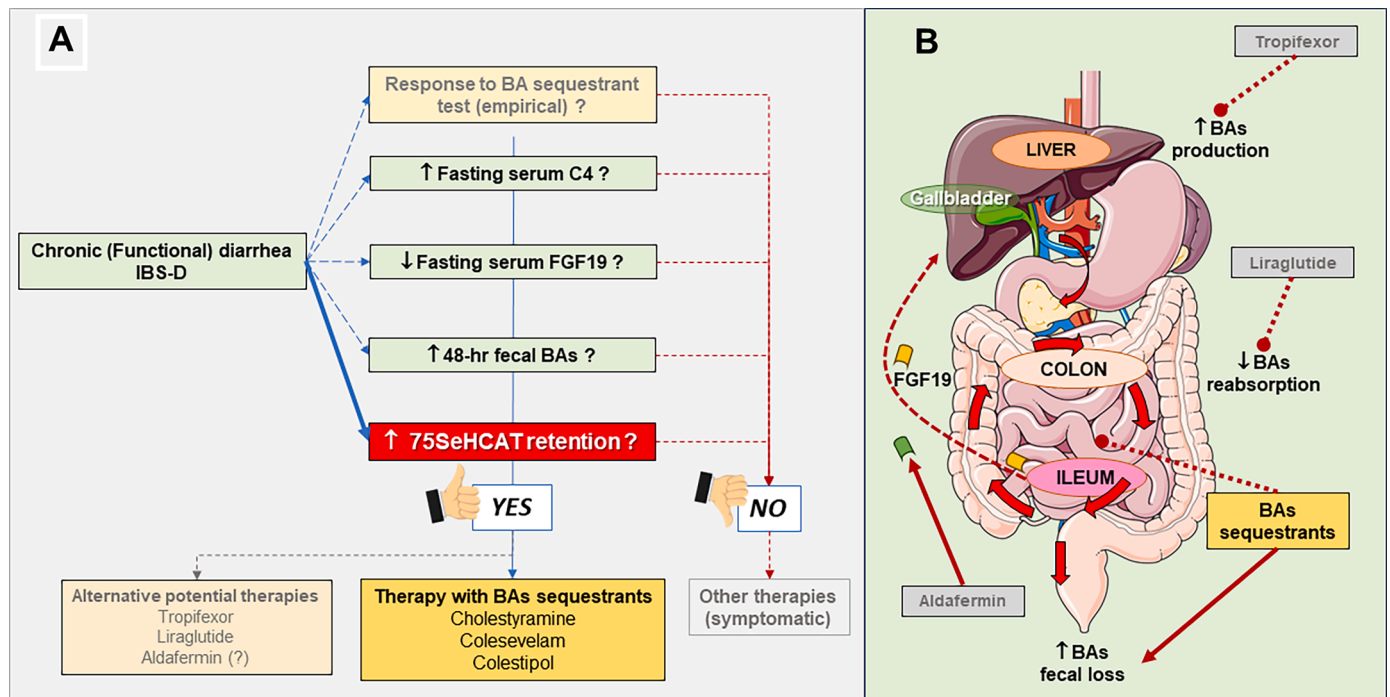


Fig. 3. Diagnosis and management of bile acid malabsorption. A) Diagnostic tests utilized in identifying patients with BAM and potential therapeutic strategies. B) Therapeutic targets with putative mechanisms of action in bile acid malabsorption (see text for details). Abbreviations: ↓, decreased; ↑, increased; BAs, bile acids; C4, 7 α -hydroxy-4-cholesten-3-one; FGF19, fibroblast growth factor 19; IBS-D, irritable bowel syndrome-diarrhoea; $^{75}\text{SeHCAT}$, Selenium HomotauroCholic Acid Test.

may lead to missed diagnosis, diagnostic delay [32], or unnecessary radiological and endoscopic examinations [33].

4.3. Fecal bile acids test

The direct measure of BAs reaching the colon and excreted in feces represents an alternative if $^{75}\text{SeHCAT}$ is not available [34–36]. The test is performed after a four-day diet containing 100 g fat followed by a 48-hour stool collection to measure the total and individual amount of fecal BAs [10,37]. The cut-off for the diagnosis of BAM is excretion of total fecal BAs in excess of 2337 $\mu\text{mol}/48$ hours or a fraction of primary BAs (CDCA and CA) exceeding 10% of the total. Another recommended cut-off is total fecal BAs in excess of 1000 $\mu\text{mol}/48$ hours plus a fraction of primary BAs exceeding 4% [36,38]. Stool frequency and consistency correlate with concentrations of primary fecal BAs [38,39]. While simple and safe, both patients and healthcare workers may find the methodology of fecal BA testing time-consuming and inconvenient. Although the determination of fecally excreted individual BA species has been previously suggested, this step is mainly reserved for experimental purposes as the simple assessment of total faecal BAs is a satisfactory option for clinical purposes.

4.4. Fasting Serum 7 α -hydroxy-4-cholesten-3-one (C4)

C4 is the precursor of hepatic BA synthesis and fasting serum C4 level is a direct measure of the activity of this pathway [10,40]. Levels of C4 >48.9 ng/mL can indicate the presence of BAM [41,42]. The measure of C4 and FGF19 (the interdependent inhibitor of BA synthesis) is validated against $^{75}\text{SeHCAT}$ [43,44]. Typical diurnal fluctuations of C4 require sampling before 9 am [10]. False-positive or false-negative results are possible in patients with cholestatic chronic liver disease, hypertriglyceridemia, and the use of statins, which can all interfere with BA synthesis [45]. C4 serum levels can change in response to factors affecting the circadian rhythm with impact on BA synthesis [46]. Due to possible confounders in the isolated diagnostic use of fasting C4 levels, combination tests are suggested for diagnostic purposes, e.g., fecal BAs

plus C4 levels [47].

4.5. Fasting serum fibroblast growth factor 19 (FGF19)

Decreased FGF19 is associated with decrease of C4 and inhibition of the conversion of cholesterol to BAs. By contrast, the diagnosis of BAM relies on decreased FGF19 and elevated C4 [48]. Serum levels of FGF19 may be used as index of ileal BA reabsorption [38,42,49]. When measured in patients with chronic diarrhoea [40,50], the cut-off value of fasting FGF19 to ascertain the diagnosis of BAM is set to <60 pg/mL [31]. As for C4 measurements, FGF19 requires sampling in the morning in the fasting subject [10,42,51].

4.6. Emerging diagnostic tests for BAM

More validation studies, including machine learning techniques, are required in this field [52,53]. Novel experimental tests are being developed in for the diagnosis of BAM, but supplemental evidence is needed. Binding of different BAs to the GPBAR-1 and FXR receptors might be analyzed by molecular imaging techniques [54], while serum lipidome profiles might distinguish individuals with BAM [53]. Detection of microbial metabolites as volatile organic compounds (VOCs) offers novel possibilities to mark gut dysbiosis in BAM, compared to healthy individuals [55–58]. This methodology is appealing and promising, but requires complex experimental procedures, standardization, and data analysis.

Current scarcity of reliable and widely available diagnostic tests represents an unmet need in cases where the diagnostic utility of BA sequestrants is insufficient to confirm the diagnosis of BAM and support the use of novel therapeutic agents in this condition.

5. Which treatment options are currently available for BAM?

The management of BAM must consider the underlying condition if the cause has been identified. In cases with unknown etiology, current therapeutic approaches include dietary modifications or BAs

Table 2
Diagnostic tests for BAM/BAD.

Test	Diagnostic accuracy	Notes
Fecal BA content	45% sensitivity, 63% specificity of detecting ⁷⁵SeHCAT <15% with primary BAs >10%; - total fecal BAs alone >2337 μmol/48 h: similar diagnostic accuracy than elevated primary BAs alone or in combination with total fecal BAs >1000 μmol/48 h in patients with BAD [10,36]	It is advisable to combine the measurement of total fecal BAs content with percentage of primary fecal bile acids to increase the diagnostic accuracy [7].
Selenium HomotauroCholic Acid Test (⁷⁵ SeHCAT)	94% sensitivity, 100% specificity in diagnosing chronic diarrhoea vs healthy subjects [147]. 87.32% sensitivity, 93.2% specificity when compared to sensitivity and specificity of serum levels of 7α-hydroxy-4-cholesten-3-one (85.2% and 71.1%, respectively) [148].	- The test with the highest accuracy [31] - The diagnostic accuracy for BAM increases with ⁷⁵SeHCAT retention rate <10% [31]
Fasting Serum 7α-hydroxy-4-cholesten-3-one (C4)	82.6% sensitivity, 84.3% specificity for detecting BAM [31,41]. - fasting serum C4 > 48.4 ng/mL: 90% sensitivity, 79% specificity, as compared to ⁷⁵SeHCAT retention value of < 10% [43] - NPV 98%, PPV 74%, as compared with ⁷⁵SeHCAT test [46]	- Combination of C4 detection with ≥1.1 daily watery (Bristol type 6–7) stools: 70% sensitivity, 95% specificity. - Combining C4, the sum of measured stool BAs, and daily watery stools increased the sensitivity to 77%, with 93% specificity [52].
Fasting Serum Fibroblast Growth Factor 19 (FGF19)	- Fasting serum FGF19 < 145 pg/mL: 58% sensitivity, 84% specificity [44]; - FGF19 <60 ng/L: 80% sensitivity, 68% specificity in chronic diarrhoea from Crohn's ileitis [149].	Biohumoral diagnosis of BAM possible with decreased FGF19 and elevated C4 levels [48].

Legend: BA, Bile acid; FGF19, fibroblast growth factor 19; ⁷⁵SeHCAT, Selenium HomotauroCholic Acid Test; C4, 7α-hydroxy-4-cholesten-3-one

sequestrants [59]. More recently, interventions to modulate gut microbiota, FGF19 and C4 levels increasingly find their role.

5.1. Dietary modifications

A low-fat diet with <20% of daily energy intake derived from fat (i.e., <30 g per day) can improve nocturnal defecation, urgency, abdominal pain, flatulence, stool consistency and frequency [60,61]. Other studies report significant improvement in abdominal pain and nocturnal defecation but no effect on stool consistency, bowel frequency, urgency, belching, flatulence, and borborygmi [61].

5.2. BA sequestrant therapy (BAST)

BAST can improve symptoms in about 50% of patients [8]. For individuals with chronic diarrhoea who have undergone ileal resection, ⁷⁵SeHCAT scan may not be essential for diagnosing BAM. Instead, it is recommended to initiate empirical BAST [9,62]. BA sequestrants include cholestyramine, colestipol, and colesevelam, and are able to improve abdominal symptoms, stool frequency, and stool consistency [63], in particular in patients with severe BAM and ⁷⁵SeHCAT retention

[32]. In addition, BAST improves scores for the “Role limitation due to physical health” dimension and in overall mental component summary, derived from the 36-Item Short Form Survey (SF-36) [19]. In an Italian retrospective study on chronic diarrhoea in 136 patients with ileal disease, cholecystectomy or post-prandial diarrhoea, BAM was confirmed in 28.1% of cases [64]. Subjects improved after 6-month treatment with cholestyramine [64]. A systematic review on 30 studies collecting 1,241 patients with established BAD and treated with BAST reported that cholestyramine was successful in 70% of the cases [65]. In a randomized clinical trial, cholestyramine (4 g twice daily for 8 weeks) vs identical hydroxypropyl cellulose sachets reduced the number of watery stools in patients with chronic diarrhoea and ⁷⁵SeHCAT 7-day retention ≤20% [66]. In another randomized trial, response rates in patients with ⁷⁵SeHCAT retention of 10% or 20% were 40% and 54%, respectively [67].

Scarce compliance can derive from unpalatability or side effects as constipation, bloating, abdominal pain, fullness, nausea, flatulence [68], and transient hypertriglyceridemia [69]. In addition, it is advisable to prescribe the administration of other medications 1 h before or 4 h after cholestyramine intake, since the resin may interfere with intestinal absorption of drugs [70] and fat-soluble vitamins [71]. Hyperchloremic metabolic acidosis is a rare possibility in patients with renal insufficiency [72,73], or in those taking aldosterone antagonists such as spironolactone [74]. This is the consequence of the resin exchanging chloride anions for BAs in the small intestine lumen.

Colesevelam has a binding affinity to BAs 4–6 times greater than cholestyramine. Furthermore, it is associated with increased patient adherence, attributed to a reduced occurrence of adverse effects such as constipation, dyspepsia, and nausea [75].

In patients with IBS-D, colesevelam notably increased total fecal BA excretion. Stool consistency improved along with a reduction in the frequency of bowel movements per week [50]. Likewise, in a randomized, double-blind, placebo-controlled trial in patients with IBS-D, colesevelam (1.875 g twice daily) improved stool consistency scores [76]. Others reported an improvement of 0.56±0.06 units per unit change in overall colonic transit at 24 h [50]. Another randomized trial enrolled 30 adults with IBS-D and evidence of increased BA synthesis or fecal excretion. Colesevelam increased the delivery of total and secondary BAs to stool, hepatic BA synthesis, and colonic mucosal expression of genes that regulate BAs, FXR and GPBAR1 [77]. In an open-label study in patients exhibiting elevated 48-hour stool BAs excretion, colesevelam improved in 10 days stool consistency and increased the excretion of fecal BAs [63]. In another study, the median number of liquid stools per week decreased from 5 to 2 [78]. Colesevelam was also effective in patients with IBS-D and decreased emptying of the ascending colon [12,50]. Colesevelam was superior to placebo at inducing remission of BAM diagnosed with C4 concentration greater than 46 ng/mL [79]. Colestipol has a better palatability, and can be a better-tolerated option than cholestyramine [80].

5.3. Farnesoid X Receptors (FXR) agonists

Pharmaceutical FXR agonists include the synthetic BA obeticholic acid (OCA) and tropifexor, a non-BA FXR agonist. Besides the feedback regulation of BA synthesis, FXR agonists decrease chloride secretion in the intestinal epithelium in response to calcium and cAMP-dependent agonists. This effect improves diarrhoea both *in vitro* and *in vivo* [81].

OCA is a potent activator of the FXR pathway, including transcriptional activation of FGF19 and inhibition of CYP7A1, the primary and rate-limiting enzyme in BA synthesis [82]. Mechanisms, safety, and symptoms in response to OCA were evaluated in an open-label, single-center pilot study lasting 6 weeks in three groups of patients with BAM/BAD. Two weeks of OCA increased FGF19 in both primary and secondary BAD, a finding due to decreased BA synthesis. OCA was well tolerated, with only a few minor adverse effects as mild headaches [82].

Tropifexor is able to inhibit BAs synthesis and colonic transit. In a

double-blind, multicenter, randomized study in patients with primary BAD, tropifexor 60 µg once daily was safe and well tolerated [83]. A recent network meta-analysis focused on randomized controlled studies that evaluated the effectiveness of different treatment options for patients with BAM. Results of seven relevant articles involving 213 participants suggested that tropifexor was the most effective treatment in raising the FGF19 levels and in reducing C4 levels, as compared to placebo [13]. Therapeutic options involving FXR agonists, however, are hardly used in clinical practice, mainly due to possible side effects (as itching during long-term therapy with OCA) and high costs.

5.4. FGF19 analogues

Aldafermin (NGM282) is the engineered analogue of recombinant human FGF19, with a 95.4% homology. In a placebo-controlled randomized trial in patients with IBS and BAM, aldafermin lowered BAS synthesis and improved stool consistency. Nevertheless, abdominal pain and stool frequency were not significantly changed [84]. Treatment-related adverse events were significantly more frequent in the aldafermin group, and four patients on aldafermin had a significant increase in low-density lipoprotein cholesterol. The utility of aldafermin in the treatment of IBS-D requires additional studies.

5.5. Glucagon-like peptide-1 (GLP-1) receptor agonists

In a randomized, double-blind, double-dummy, non-inferiority clinical trial, patients with BAM diagnosed by ⁷⁵SeHCAT were randomly assigned to liraglutide (one daily subcutaneous injection uptitrated from 0.6–1.8 mg per day over 3 weeks) or colesevelam (three capsules of 625 mg twice daily) for 6 weeks [30]. Results showed a significant risk difference of –27% in favour of liraglutide in reducing daily stool frequency. Mild nausea was reported by 6 subjects in the liraglutide group, and 1 subject in the colesevelam group. Liraglutide likely allows a greater degree of passive and active BAs reabsorption through the small intestine, a mechanism leading to reduced spillover of BAs [77,85]. In addition, in a network meta-analysis, liraglutide compared to colesevelam and placebo more efficiently decreased fecal BA concentration [13].

5.6. Modulation of gut microbiota

Dysbiosis may contribute to excessive BA excretion, as shown in patients with IBS-D [3].

Thus, modulation of gut microbiota may improve BAM [38,49]. A decreased abundance of the *Ruminococcaceae* genera [86] and an increase in *Clostridia* bacteria have been reported in patients with IBS-D. Microbial changes correlate with fecal BAs and serum C4 [3]. Fermented soymilk may alleviate diarrhoea by decreasing harmful bacteria and stimulating the growth of healthy species while increasing the levels and anti-inflammatory effects of ursolic acid [87]. However, more evidence is required when considering the benefits of gut microbiota-modulating therapies in BAM.

5.7. Comparative pharmaceutical trials

A recent systematic review and network meta-analysis compared the efficacy of cholestyramine, colesevelam, tropifexor, loperamide, and liraglutide in managing BAM based on effectiveness outcomes that included daily bowel movement frequency, stool type according to Bristol Stool Form Scale, serum levels of FGF19 and C4, and the total amount of fecal BAs [13]. Tropifexor was the most effective medication in increasing FGF19 levels and reducing C4 levels, while liraglutide was the most potent treatment for reducing fecal BAs concentrations. The superiority of liraglutide in decreasing fecal BAs concentration vs. colesevelam and placebo can represent a valid alternative in patients with BAM, although further evidence is required [13].

6. Conclusions

The possibility of BAM in patients with unexplained, chronic diarrhoea requires awareness, especially in presence of ileal disease or after gastrointestinal surgery. While a great deal of the pathogenic mechanisms leading to BAM have been unravelled, the diagnosis and management of this condition remain unsatisfactory. Most cases of BAM are empirically affirmed based on their clinical response to BA sequestrants. However, this diagnostic approach is fraught by suboptimal accuracy. By contrast, the highly specific ⁷⁵SeHCAT test is still mostly unavailable for routine use. Other tests as C4 and FGF19 serum levels and measurement of fecal BA excretion still have limited use in clinical practice. Thus, further efforts are needed to obtain widely accessible, low-cost diagnostic tools. Recent advances in the pathophysiology of BAM point to the involvement of specific receptors as FXR, GPBAR-1, and FGFR4, and hold the promise of improved therapeutic approaches for this disorder. Future trials will hopefully identify safe and well-tolerated forms of controlling BAM to improve the quality of life of affected patients.

Author contributions

M.K., A.D.C. and P.P. conceived the idea for the manuscript and wrote the draft; P.P. conceived the figures; G.B. and P.P. reviewed the manuscript.

Conflict-of-interest statement

All the authors report having no relevant conflicts of interest for this article.

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